

Answers of Application Build and Release Automation Lab Assignment

Question1: Linux Project Setup

5 Marks

Create:

<FirstName>-GLA-HealthSystem

Inside the directory create:

- src
- docs
- config
- backup

Create:

health_info.txt

The file must contain:

- Student Name
- Roll Number
- Subject Name
- Project Name
- Semester

Demonstrate appropriate commands to:

- Display current directory.
- Navigate through directories.
- List files.
- Display file contents.
- Display the directory structure.

Answer 1:

```

Last login: Tue Sep 1 16:41:16 on ttynd0
paridishakla@Paridish-Macbook-Air ~ % pwd
/Users/paridishakla
paridishakla@Paridish-Macbook-Air ~ % mkdir Paridih-QIA-HealthSystem
paridishakla@Paridish-Macbook-Air ~ % ls
2.py                               aws_files.py
3.py                               b.txt
3.python                           bucket.py
4.py                               c.txt
Applications                       cloud.py
BankAccount.Classes               cloudaws.py
BankAccount.Java                  code of user_data1.ipynb
BinarySearch.Classes              connect.py
BinarySearch.Java                 create_bucket.py
BubbleSort.java                   data_process.py
Cloudwatch.py                     domsProject.html
Desktop                            domsProject.html
Documents                          directory_Paridih
Downloads                          doc.py
Git_Practical                     html
Helloworld                         item
Helloworld.G                       item.java
Helloworld.GVM                    java_UnsortedLinkedList2
JavaProjects                       lambda_function.py
Library                            learn.py
Movies                             learnSearch.java
Music                              list_bucket.py
Paridih-QIA-CourseRegistration     main
Paridih-QIA-HealthSystem           main.c
Paridih-Bucket-789.py              kafka-kafka-deploy-qaests
Pictures                           mynewbucket-9999.py
Program                             mynew
Program2.G                         mynew.html
Program2.GVM                       new-ec2-file.py
Proj1.java                         part
Proj2.java                         paridih-unique-bucket-9999.py
Public                             paridih.html
Second                             portfolio
Second.G                           project.py
Second.GVM                         public class UnsortedLinkedList2
UnsortedLinkedList                 python
UnsortedLinkedList.java            python.py
UnsortedLinkedList2.java           xl_exploiter.py
Virtual_Machines_Localized         xl_exploiter.py
VirtualBox.WG                      sort.class
a.txt                              sort.java
b.txt                              sorted
b.py                               tempCodeRunnerFile.js
cite.html                          test
aws-ec2-practice.py               test
aws-iam-practice.py               test
paridishakla@Paridish-Macbook-Air ~ % cd Paridih-QIA-HealthSystem
paridishakla@Paridish-Macbook-Air ~ % cd Paridih-QIA-HealthSystem % pwd
/Users/paridishakla/Paridih-QIA-HealthSystem
paridishakla@Paridish-Macbook-Air ~ % cd Paridih-QIA-HealthSystem % mkdir src docs config backup
paridishakla@Paridish-Macbook-Air ~ % cd Paridih-QIA-HealthSystem % ls
Backup    config    docs     health_info.txt src
paridishakla@Paridish-Macbook-Air ~ % cd Paridih-QIA-HealthSystem % nano health_info.txt
Student Name: Paridish Shukla
Roll Number: 241581892
Subject Name: Application Build and Release Automation Lab
Project Name: QIA HealthTech Campus Health Appointment System

```

```

Semester: 6th Semester
paridishakla@Paridish-Macbook-Air ~ % cd Paridih-QIA-HealthSystem % cd src
paridishakla@Paridish-Macbook-Air ~ % cd Paridih-QIA-HealthSystem % cd src
/Users/paridishakla/Paridih-QIA-HealthSystem/src
paridishakla@Paridish-Macbook-Air ~ % cd ..
paridishakla@Paridish-Macbook-Air ~ % cd Paridih-QIA-HealthSystem % pwd
/Users/paridishakla/Paridih-QIA-HealthSystem
paridishakla@Paridish-Macbook-Air ~ % cd Paridih-QIA-HealthSystem % ls
Backup    config    docs     health_info.txt src
paridishakla@Paridish-Macbook-Air ~ % cd Paridih-QIA-HealthSystem % sudo apt install tree
Password:
The operation couldn't be completed. Unable to locate a Java Runtime.
Please visit http://www.java.com for information on installing Java.

paridishakla@Paridish-Macbook-Air ~ % cd Paridih-QIA-HealthSystem % tree
src:
├── command_not_found.py
├── paridishakla@Paridish-Macbook-Air ~ % cd Paridih-QIA-HealthSystem % brew --version
Homebrew 6.8.18
paridishakla@Paridish-Macbook-Air ~ % cd Paridih-QIA-HealthSystem % brew install tree
... Auto-updating Homebrew...
Adjust how often this is run with 'BREW_UPDATE_SECS' or disable with
'SHOWBREW_NO_AUTO_UPDATE=1'. Hide these hints with 'SHOWBREW_NO_HINTS=1' (see 'man brew').
... Auto-updated Homebrew!
Updated 2 taps (homebrew/core and homebrew/cask).

... New Formulae
amy: Nostr client for the Aemthyst project
betterplotlake: Rewrote Globe key for faster input source switching
budget-tracker: Feature rich TUI budget tracker app
delf-wi: Local searchable memory over the session histories of coding agents
fmgpg80: Play, record, convert, and stream mp3net audio and video codecs
gffprocessor: Compare, merge and annotate GFF/GTF/Transcript files
gicli: Run Gitlab CI/CD pipelines locally
jodex: AI coding agent harness for the terminal
leaves: Text-mode disk usage visualization utility
libmysofa: Reader for AES SQA files to get better MP3s
linuxcast: Weather, tides, the sun, the moon, and maps, drawn for the terminal
mysql-client@9.7: Open source relational database management system
pulsarui: Pulsar client. Easy to search, browse, and download artworks
pLink-ng: Whole-genome association analysis toolset (PLINK 2.0)
pychibi: Monte Carlo event generator
readwise-cli: Command-line interface for Readwise and Reader
robot-framework-toboco: Static code analysis tool (linter) and code formatter for Robot Framework
robotcode: Ultimate Robot Framework Toolset
rubocopi: Ruby static code analyzer and formatter, based on the community Ruby style guide
sabotai: Fast, safe-by-design, driverless, Win-first DB TUI with ER diagrams
snoopi: Persistent shell session manager
skillspector: Security scanner for AI agent skills
speedtest-go: CLI and Go API to test Internet Speed using speedtest.net
telnet: Netcat-like tool over Telnet's data plane, without its control plane
tele: Keyboard-first Telegram client for the terminal, written in Go
tiroq: Trigram-indexed grep for fast regex search in large codebases
virgintender: VIOU, virtual OpenAI, tenders
vevd: Very fast Essential Video Decoder, VP8-5 EVC (Essential Video Coding)
vevd: Very fast Essential Video Encoder, VP8-5 EVC (Essential Video Coding)

... New Casks
afterglow: Classic After Dark screen savers emulator
agentlao: IDE for agent-based development
airstats: Menu bar system monitor
amethyst-nostr: Nostr client
analog: AI notepad for meetings
android-performance-analyzer: Toolchain for profiling apps and games
arm-performix: Performance analysis toolkit for Arm server and cloud environments
atlas-apps: Source control for coding agents
bifrost: Samsung firmware downloader

```

```
bramble: Password manager
channel-works: AI Business OS for customer support, analytics, collaboration, and marketing
clawed-on-desk: Desktop pet that reacts to AI coding agents
crisp: Menu bar display manager: IDC brightness, HDR?, presets, virtual displays
davit: GUI for Apple's container CLI
du-pro: Query, explore, and manage your databases with built-in AI
diskboard: Disk benchmark and S.M.A.R.T. health monitor
dockdoorpro: Dock replacement with widgets, profiles and window previews
droopy: Drag and drop file shelf
font-asap-sharp
font-clarity-city
font-isoetra
font-kriya
font-libre-caslon-condensed
font-nevermind
font-paper-mono
font-valley-sans
gamenaker@beta: Complete development tool for making 2D games
keychron-assistant: Companion app for Keychron Launcher's Quick Start feature
leipad: Game network accelerator
mac-performance-monitor: Menu bar performance monitor with recorded history and analytics
macshot-offline: Screenshot and screen recording tool without upload integrations
mangodisk: Disk cleaner and storage analyser
markviewer: Minimal markdown editor
minmaxcal: Minimal menu bar calendar, maxinal full-screen notifications
motrix@beta: Open-source download manager
native: Run AI models locally
onion: Tiling window manager
pixecake: AI photo editing software for commercial photography
quarkcloudrives: Cloud storage and file management platform
riverscript: AI platform for recording and transcribing system audio from any app
sintel: Minecraft launcher built with the community
skills-manager: Manage, sync, and organise AI agent skills across coding tools
switch: Tree-based app switcher
tabularis: Lightweight database management tool
tcp-viewer: Packet capture and inspection tool
terminal-browser: Terminal-based web browser
thoughttag: Visual workspace for editable LLM context graphs
token-monitor: Monitor token usage, costs, and limits across AI coding tools
tone3888: Amp modeling plug-in for Neural Amp Modeler captures and impulse responses
triggerflo: Focus timer and Kanban board for tracking tasks
uniloth: Desktop application for Uniloth Studio
vernier-graphical-analysis: Instrument data analysis tool
waku: Native desktop app for coding agents

You have 13 outdated formulae and 2 outdated casks installed.

Warning: The following taps are not trusted:
  derailed/k9s
  ubuntu/microk8s

Homebrew is currently ignoring formulae, casks and commands from these taps because tap trust is required.

Prefer trusting only the specific formulae, casks or commands you need.
Trust installed formulae from these taps with:
  brew trust --formula derailed/k9s/k9s
  brew trust --formula ubuntu/microk8s/microk8s
Trust other specific casks and commands with:
  brew trust --cask <user>/<tap>/<cask>
  brew trust --command <user>/<tap>/<command>
Whole-tap trust is broader and includes all current and future formulae,
casks and commands from the listed taps. Trust whole taps with:
  brew trust derailed/k9s ubuntu/microk8s
Untap then with:
  brew untap derailed/k9s ubuntu/microk8s
```

```
markviewer: Minimal markdown editor
minmaxcal: Minimal menu bar calendar, maxinal full-screen notifications
motrix@beta: Open-source download manager
native: Run AI models locally
onion: Tiling window manager
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  brew trust --command <user>/<tap>/<command>
Whole-tap trust is broader and includes all current and future formulae,
casks and commands from the listed taps. Trust whole taps with:
  brew trust derailed/k9s ubuntu/microk8s
Untap then with:
  brew untap derailed/k9s ubuntu/microk8s

To disable trust checks:
  export HOMEBREW_NO_REQUIRE_TAP_TRUST=1
This is not recommended and will be removed in a later release.
For more information, see:
  https://docs.brew.sh/FAQ#trust
  - Downloading bottle manifests
    * Bottle Manifest tree (2.3.2) Downloaded 7.5KB/ 7.5KB
  - Would install 1 formula:
    tree
  - Patching downloads for: tree
    * Bottle tree (2.3.2) Downloaded 66.1KB/ 66.1KB
  - Pouring tree-2.3.2.arm64_sonoma.bottle.tar.gz
    * /opt/homebrew/Cellar/tree/2.3.2: 9 files, 195.3KB
paridishishu@Paridish-MacBook-Air Paridish-GUI-health$ system N tree
.
├── backup
├── config
├── docs
├── health_info.txt
├── src
└──

5 directories, 1 file
paridishishu@Paridish-MacBook-Air Paridish-GUI-health$ system N
```

Question 2: Git Version Control

6 Marks

1. Initialize Git inside the project directory.
2. Configure Git identity if required.
3. Check Git status.
4. Stage the project files.
5. Create an initial commit.
6. Modify `health_info.txt` by adding:
 - o Project objective
 - o Student's program/course
7. View the changes made.
8. Check Git status.
9. Stage the modification.
10. Create a second commit.
11. Display the commit history.

Ans 2:

```
paridhishukla@Paridhi-MacBook-Air Paridhi-GLA-HealthSystem % pwd
/Users/paridhishukla/Paridhi-GLA-HealthSystem
paridhishukla@Paridhi-MacBook-Air Paridhi-GLA-HealthSystem % git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: will change to 'main' in Git 3.0. To configure the initial branch name
hint: to use in all of your new repositories, which will suppress this warning,
hint: call:
hint:   git config --global init.defaultBranch <name>
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:   git branch -m <name>
hint: Disable this message with 'git config set advice.defaultBranchName false'
Initialized empty Git repository in /Users/paridhishukla/Paridhi-GLA-HealthSystem/.git/
paridhishukla@Paridhi-MacBook-Air Paridhi-GLA-HealthSystem % git config --global user.name Paridhi
paridhishukla@Paridhi-MacBook-Air Paridhi-GLA-HealthSystem % git config --global user.email p.shukla22@gmail.com
paridhishukla@Paridhi-MacBook-Air Paridhi-GLA-HealthSystem % git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    health_info.txt

nothing added to commit but untracked files present (use "git add" to track)
paridhishukla@Paridhi-MacBook-Air Paridhi-GLA-HealthSystem % git add .
paridhishukla@Paridhi-MacBook-Air Paridhi-GLA-HealthSystem % git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
    new file:   health_info.txt

paridhishukla@Paridhi-MacBook-Air Paridhi-GLA-HealthSystem % git commit -m "Initial project setup"
[master (root-commit) f009978] Initial project setup
 1 file changed, 5 insertions(+)
 create mode 100644 health_info.txt
paridhishukla@Paridhi-MacBook-Air Paridhi-GLA-HealthSystem % git status
On branch master
nothing to commit, working tree clean
paridhishukla@Paridhi-MacBook-Air Paridhi-GLA-HealthSystem % nano health_info.txt
paridhishukla@Paridhi-MacBook-Air Paridhi-GLA-HealthSystem % git diff
diff --git a/health_info.txt b/health_info.txt
index f1934de..46adb23 100644
--- a/health_info.txt
+++ b/health_info.txt
@@ -3,0 @@ Roll Number: 2115091892
Subject Name: Application Build and Release Automation Lab
Project Name: GLA HealthTech Campus Health Appointment System
Semester: 5th Semester
Project Objective: To develop a Campus Health Appointment System for efficient management and booking of health appointments.
<Student Program/Course: B.Tech Computer Science and Engineering
paridhishukla@Paridhi-MacBook-Air Paridhi-GLA-HealthSystem % git status
On branch master
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
```



```

On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
    new file:   health_info.txt

paridhishukla@Paridhi-MacBook-Air Paridhi-GLA-HealthSystem % git commit -m "Initial project setup"
[master (root-commit) f089978] Initial project setup
 1 file changed, 5 insertions(+)
 create mode 100644 health_info.txt
paridhishukla@Paridhi-MacBook-Air Paridhi-GLA-HealthSystem % git status
On branch master
nothing to commit, working tree clean
paridhishukla@Paridhi-MacBook-Air Paridhi-GLA-HealthSystem % nano health_info.txt
paridhishukla@Paridhi-MacBook-Air Paridhi-GLA-HealthSystem % git diff
diff --git a/health_info.txt b/health_info.txt
index f9934de..6d0d23 100644
--- a/health_info.txt
+++ b/health_info.txt
@@ -3,3 +3,6 @@ Roll Number: 2415081892
 Subject Name: Application Build and Release Automation Lab
 Project Name: GLA HealthTech Campus Health Appointment System
 Semester: 5th Semester
+Project Objective: To develop a Campus Health Appointment System for efficient management and booking of health appointments.
+
+Student Program/Course: B.Tech Computer Science and Engineering
paridhishukla@Paridhi-MacBook-Air Paridhi-GLA-HealthSystem % git status
On branch master
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
    modified:   health_info.txt

no changes added to commit (use "git add" and/or "git commit -a")
paridhishukla@Paridhi-MacBook-Air Paridhi-GLA-HealthSystem % git add health_info.txt
paridhishukla@Paridhi-MacBook-Air Paridhi-GLA-HealthSystem % git status
On branch master
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
    modified:   health_info.txt

paridhishukla@Paridhi-MacBook-Air Paridhi-GLA-HealthSystem % git commit -m "Update health system information"
[master f17bd89] Update health system information
 1 file changed, 3 insertions(+)
paridhishukla@Paridhi-MacBook-Air Paridhi-GLA-HealthSystem % git log --oneline
f17bd89 (HEAD -> master) Update health system information
f089978 Initial project setup
paridhishukla@Paridhi-MacBook-Air Paridhi-GLA-HealthSystem % git log
commit f17bd89d9f90a8633f89d76713b17ec06f519 (HEAD -> master)
Author: Paridhi cp.shukla210@gmail.com
Date:   Mon Sep 7 12:05:58 2026 +0530

    Update health system information

commit f089978a494c3cfd82a29fc75c3d0dfe47ec85
Author: Paridhi cp.shukla210@gmail.com
Date:   Mon Sep 7 11:58:53 2026 +0530

    Initial project setup
paridhishukla@Paridhi-MacBook-Air Paridhi-GLA-HealthSystem % git status
On branch master
nothing to commit, working tree clean
paridhishukla@Paridhi-MacBook-Air Paridhi-GLA-HealthSystem %

```

Question3: Independent Feature Development

7 Marks

The team wants to develop an **Appointment Module** independently.

1. Create a branch named:

appointment-feature

2. Switch to the branch.
3. Inside **src**, create:

appointment.txt

4. Add suitable content describing:
 - o Purpose of the Appointment Module
 - o At least three features
 - o Expected users of the module
5. Stage and commit the file.
6. Display all branches.
7. Display the current branch.
8. Return to the main branch.
9. Demonstrate that the feature work was performed separately.

Ans 3:

```
paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-HealthSystem % git branch appointment-feature
paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-HealthSystem % git switch appointment-feature
Switched to branch 'appointment-feature'
paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-HealthSystem % git branch
* appointment-feature
  master
paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-HealthSystem % cd src
paridhishukla@Paridhis-MacBook-Air src % pwd
/Users/paridhishukla/Paridhi-GLA-HealthSystem/src
paridhishukla@Paridhis-MacBook-Air src % touch appointment.txt
paridhishukla@Paridhis-MacBook-Air src % ls
appointment.txt
paridhishukla@Paridhis-MacBook-Air src % nano appointment.txt
paridhishukla@Paridhis-MacBook-Air src % cat appointment.txt
Appointment Module

Purpose:
The Appointment Module is designed to allow users to book and manage health appointments efficiently through the Campus Health Appointment System.

Features:
1. Book a health appointment.
2. View scheduled appointments.
3. Cancel or manage an existing appointment.

Expected Users:
The expected users of this module are students, faculty members, and campus health center staff.

paridhishukla@Paridhis-MacBook-Air src % git status
On branch appointment-feature
Untracked files:
  (use "git add <file>..." to include in what will be committed)
  .

nothing added to commit but untracked files present (use "git add" to track)
paridhishukla@Paridhis-MacBook-Air src % cd .
paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-HealthSystem % pwd
/Users/paridhishukla/Paridhi-GLA-HealthSystem
paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-HealthSystem % git add src/appointment.txt
paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-HealthSystem % git status
On branch appointment-feature
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
    new file:   src/appointment.txt

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    src/appointment.txt

paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-HealthSystem % cat src/appointment.txt
paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-HealthSystem % cat src/appointment.txt
Appointment Module

Purpose:
The Appointment Module is designed to allow users to book and manage health appointments efficiently through the Campus Health Appointment System.

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1. Book a health appointment.
2. View scheduled appointments.
3. Cancel or manage an existing appointment.

Expected Users:
The expected users of this module are students, faculty members, and campus health center staff.

paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-HealthSystem % rm src/appointment.txt
paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-HealthSystem % ls src
```

```
paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-HealthSystem % git status
On branch appointment-feature
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
    new file:   src/appointment.txt

Changes not staged for commit:
  (use "git add/rm <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
    deleted:    src/appointment.txt

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    src/appointment.txt

paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-HealthSystem % git restore --staged src/appointment.txt
paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-HealthSystem % git status
On branch appointment-feature
Untracked files:
  (use "git add <file>..." to include in what will be committed)
    src/

nothing added to commit but untracked files present (use "git add" to track)
paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-HealthSystem % git add src/appointment.txt
paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-HealthSystem % git status
On branch appointment-feature
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
    new file:   src/appointment.txt

paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-HealthSystem % git commit -m "Add appointment module"
[appointment-feature 4fcfe7b] Add appointment module
 2 file changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 src/appointment.txt
paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-HealthSystem % git status
On branch appointment-feature
nothing to commit, working tree clean
paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-HealthSystem % git branch
* appointment-feature
  master
paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-HealthSystem % git branch --show-current
appointment-feature
paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-HealthSystem % git log --oneline --all --decorate
4fcfe7b (HEAD -> appointment-feature) Add appointment module
f17bd89 (master) Update health system information
f809978 Initial project setup
paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-HealthSystem % git switch master
Switched to branch 'master'
paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-HealthSystem % git status
On branch master
nothing to commit, working tree clean
paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-HealthSystem % ls src
ls: src: No such file or directory
paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-HealthSystem % git branch --show-current
master
paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-HealthSystem % git ls-tree -r appointment-feature --name-only
health.info.txt
src/appointment.txt
paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-HealthSystem % git ls-tree -r master --name-only
health.info.txt
paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-HealthSystem % git branch --show-current
master
paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-HealthSystem % git ls-tree -r appointment-feature --name-only
health.info.txt
src/appointment.txt
```

Question4: Merge and Resolve Conflict

8 Marks

Part A – Merge Feature

Merge:

appointment-feature

into the main branch.

Verify that `appointment.txt` is now available.

Part B – Conflict

1. Create a branch:

system-update

2. Modify a particular line in `health_info.txt`.
3. Commit the change.
4. Return to the main branch.
5. Modify the **same line differently**.
6. Commit the change.
7. Attempt to merge `system-update`.
8. Identify the conflict.
9. Open the file and identify the Git conflict markers.
10. Resolve the conflict manually.
11. Remove the conflict markers.
12. Save the final version.
13. Stage the resolved file.
14. Commit the resolved version.
15. Verify the final Git status and commit history.

Ans 4:

```
paridishukla@Paridhis-MacBook-Air Paridhi-GLA-HealthSystem % git switch -c system-update
fatal: a branch named 'system-update' already exists
paridishukla@Paridhis-MacBook-Air Paridhi-GLA-HealthSystem % git branch
  appointment-feature
* master
paridishukla@Paridhis-MacBook-Air Paridhi-GLA-HealthSystem % nano health_info.txt
paridishukla@Paridhis-MacBook-Air Paridhi-GLA-HealthSystem % cat health_info.txt
Student Name: Paridhi Shukla
Roll Number: 2415001892
Subject Name: Application Build and Release Automation Lab
Project Name: GLA HealthTech Campus Health Appointment System
Semester: 5th Semester
Project Objective: To provide students with a simple and efficient system for booking campus health appointments.
Student Program/Course: B.Tech Computer Science and Engineering
paridishukla@Paridhis-MacBook-Air Paridhi-GLA-HealthSystem % git diff
diff --git a/health_info.txt b/health_info.txt
index 46e6b22..d74eeed 100644
+++ b/health_info.txt
@@ -3,6 +3,6 @@ Roll Number: 2415001892
 Subject Name: Application Build and Release Automation Lab
 Project Name: GLA HealthTech Campus Health Appointment System
 Semester: 5th Semester
-Project Objective: To develop a Campus Health Appointment System for efficient management and booking of health appointments.
+Project Objective: To provide students with a simple and efficient system for booking campus health appointments.
 Student Program/Course: B.Tech Computer Science and Engineering
paridishukla@Paridhis-MacBook-Air Paridhi-GLA-HealthSystem % git add health_info.txt
paridishukla@Paridhis-MacBook-Air Paridhi-GLA-HealthSystem % git status
On branch system-update
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        modified:   health_info.txt
paridishukla@Paridhis-MacBook-Air Paridhi-GLA-HealthSystem % git commit -m "Update project objective"
[system-update c2d7521] Update project objective
1 file changed, 1 insertion(+), 2 deletions(-)
paridishukla@Paridhis-MacBook-Air Paridhi-GLA-HealthSystem % git status
On branch system-update
nothing to commit, working tree clean
paridishukla@Paridhis-MacBook-Air Paridhi-GLA-HealthSystem % cat health_info.txt
Student Name: Paridhi Shukla
Roll Number: 2415001892
Subject Name: Application Build and Release Automation Lab
Project Name: GLA HealthTech Campus Health Appointment System
Semester: 5th Semester
Project Objective: To provide students with a simple and efficient system for booking campus health appointments.
Student Program/Course: B.Tech Computer Science and Engineering
paridishukla@Paridhis-MacBook-Air Paridhi-GLA-HealthSystem % nano health_info.txt
paridishukla@Paridhis-MacBook-Air Paridhi-GLA-HealthSystem % git diff
diff --git a/health_info.txt b/health_info.txt
index d74eeed..2b1eb9f 100644
+++ b/health_info.txt
@@ -3,6 +3,6 @@ Roll Number: 2415001892
 Subject Name: Application Build and Release Automation Lab
 Project Name: GLA HealthTech Campus Health Appointment System
 Semester: 5th Semester
-Project Objective: To provide students with a simple and efficient system for booking campus health appointments.
+Project Objective: To enable students and staff to easily schedule and manage appointments at the campus health center.
 Student Program/Course: B.Tech Computer Science and Engineering
paridishukla@Paridhis-MacBook-Air Paridhi-GLA-HealthSystem % git add health_info.txt
paridishukla@Paridhis-MacBook-Air Paridhi-GLA-HealthSystem % git status
On branch system-update
```



```

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
    modified:   health_info.txt

paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-HealthSystem % git commit -m "Update project objective on master"
[system-update ba28a37] Update project objective on master
 1 file changed, 1 insertion(+), 1 deletion(-)
paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-HealthSystem % git merge system-update
Already up to date.
paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-HealthSystem % git switch master
Switched to branch 'master'
paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-HealthSystem % cat health_info.txt
Student Name: Paridhi Shukla
Roll Number: 2415001892
Subject Name: Application Build and Release Automation Lab
Project Name: GLA HealthTech Campus Health Appointment System
Semester: 5th Semester
Project Objective: To develop a Campus Health Appointment System for efficient management and booking of health appointments.

Student Program/Course: B.Tech Computer Science and Engineering
paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-HealthSystem % nano health_info.txt
paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-HealthSystem % git branch --show-current
master
paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-HealthSystem % git log --oneline --all --decorate
ba28a37 (system-update) Update project objective on master
cc2f192 Update project objective
4fcfe7b (HEAD -> master, appointment-feature) Add appointment module
f17bd09 Update health system information
f009978 Initial project setup
paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-HealthSystem % nano health_info.txt
paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-HealthSystem % git switch system-update
error: Your local changes to the following files would be overwritten by checkout:
  health_info.txt
Please commit your changes or stash them before you switch branches.
Aborting
paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-HealthSystem % git reset --hard cc2f192
HEAD is now at cc2f192 Update project objective
paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-HealthSystem % git log --oneline --all --decorate
ba28a37 (system-update) Update project objective on master
cc2f192 (HEAD -> master) Update project objective
4fcfe7b (appointment-feature) Add appointment module
f17bd09 Update health system information
f009978 Initial project setup
paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-HealthSystem % cat health_info.txt
Student Name: Paridhi Shukla
Roll Number: 2415001892
Subject Name: Application Build and Release Automation Lab
Project Name: GLA HealthTech Campus Health Appointment System
Semester: 5th Semester
Project Objective: To provide students with a simple and efficient system for booking campus health appointments.
Student Program/Course: B.Tech Computer Science and Engineering
paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-HealthSystem % nano health_info.txt
paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-HealthSystem % git diff
diff --git a/health_info.txt b/health_info.txt
index d74cead..4a9ee6e 100644
--- a/health_info.txt
+++ b/health_info.txt
@@ -3,5 +3,5 @@ Roll Number: 2415001892
 Subject Name: Application Build and Release Automation Lab
 Project Name: GLA HealthTech Campus Health Appointment System
 Semester: 5th Semester
-Project Objective: To provide students with a simple and efficient system for booking campus health appointments.
+Project Objective: To help students easily schedule and manage appointments at the campus health center.
Student Program/Course: B.Tech Computer Science and Engineering
paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-HealthSystem % git add health_info.txt

```

```

paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-HealthSystem % git add health_info.txt
paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-HealthSystem % git status
On branch master
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
    modified:   health_info.txt

paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-HealthSystem % git commit -m "Update project objective on master"
[master b8b69c8] Update project objective on master
 1 file changed, 1 insertion(+), 1 deletion(-)
paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-HealthSystem % git status
On branch master
nothing to commit, working tree clean
paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-HealthSystem % git merge system-update
Auto-merging health_info.txt
CONFLICT (content): Merge conflict in health_info.txt
Automatic merge failed; fix conflicts and then commit the result.
paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-HealthSystem % cat health_info.txt
Student Name: Paridhi Shukla
Roll Number: 2415001892
Subject Name: Application Build and Release Automation Lab
Project Name: GLA HealthTech Campus Health Appointment System
Semester: 5th Semester
<<<<<< HEAD
Project Objective: To help students easily schedule and manage appointments at the campus health center.
=====
Project Objective: To enable students and staff to easily schedule and manage appointments at the campus health center.
>>>>>> system-update
Student Program/Course: B.Tech Computer Science and Engineering
paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-HealthSystem % nano health_info.txt
paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-HealthSystem % git status
On branch master
You have unmerged paths.
  (fix conflicts and run "git commit")
  (use "git merge --abort" to abort the merge)

Unmerged paths:
  (use "git add <file>..." to mark resolution)
    both modified:   health_info.txt

no changes added to commit (use "git add" and/or "git commit -a")
paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-HealthSystem % git add health_info.txt
paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-HealthSystem % git status
On branch master
All conflicts fixed but you are still merging.
  (use "git commit" to conclude merge)

Changes to be committed:
  modified:   health_info.txt

paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-HealthSystem % git commit -m "Resolve merge conflict"
[master 4b6c49c] Resolve merge conflict
paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-HealthSystem % git status
On branch master
nothing to commit, working tree clean
paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-HealthSystem % git log --oneline --graph --all
* 4b6c49c (HEAD -> master) Resolve merge conflict
/
* ba28a37 (system-update) Update project objective on master
* b8b69c8 Update project objective on master
/
* cc2f192 Update project objective
* 4fcfe7b (appointment-feature) Add appointment module
* f17bd09 Update health system information
* f009978 Initial project setup

```


Question 5:Repository Demonstration

4 Marks

Demonstrate:

1. Project directory structure.
2. Git repository.
3. Commit history.
4. Feature branch.
5. Successful merge.
6. Merge conflict and resolution.

Marks Distribution

Area	Marks
Linux project setup	5
Git version control	6
Branching and feature development	7
Merge and conflict resolution	8
Final demonstration	4
Total	30

Ans 5:

```

paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-HealthSystem % git status
On branch master
All conflicts fixed but you are still merging.
(use "git commit" to conclude merge)

Changes to be committed:
  modified:   health_info.txt

paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-HealthSystem % git commit -m "Resolve merge conflict"
[master 4b6c49c] Resolve merge conflict
paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-HealthSystem % git status
On branch master
nothing to commit, working tree clean
paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-HealthSystem % git log --oneline --graph --all
* 4b6c49c (HEAD -> master) Resolve merge conflict
|
| * ba28a37 (system-update) Update project objective on master
| * | b8b69c8 Update project objective on master
| /
| * cc2f192 Update project objective
| * 4fcfe7b (appointment-feature) Add appointment module
| * f17bd89 Update health system information
| * f889978 Initial project setup
paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-HealthSystem % pwd
/Users/paridhishukla/Paridhi-GLA-HealthSystem
paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-HealthSystem % find . -maxdepth 2 -print
.
./config
./docs
./git
./git/ORIG_HEAD
./git/config
./git/objects
./git/HEAD
./git/info
./git/logs
./git/description
./git/hooks
./git/refs
./git/index
./git/COMMIT_EDITMSG
./health_info.txt
./backup
./src
./src/appointment.txt
paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-HealthSystem % git status
On branch master
nothing to commit, working tree clean
paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-HealthSystem % git log --oneline --graph --all
* 4b6c49c (HEAD -> master) Resolve merge conflict
|
| * ba28a37 (system-update) Update project objective on master
| * | b8b69c8 Update project objective on master
| /
| * cc2f192 Update project objective
| * 4fcfe7b (appointment-feature) Add appointment module
| * f17bd89 Update health system information
| * f889978 Initial project setup
paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-HealthSystem % git branch
  appointment-feature
* master
  system-update
paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-HealthSystem % git branch --show-current
master
paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-HealthSystem %

```